Amir Mirzai Golpayegani

Calgary, AB, Canada

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linkedin.com/in/amir-golpayegani-218148165 | github.com/amirgolpaa24 | Portfolio

Summary

M.Sc. Computer Science graduate in Calgary with experience building Python systems for backend services, data processing, machine learning workflows, and scientific computing. Skilled in Python, Django, PostgreSQL, Linux, Git, Docker, REST APIs, and scalable data pipelines, with hands-on experience developing reliable software for large geospatial datasets and AI workflow automation.

Education

M.Sc. in Computer Science GPA: **3.94**/4.0 Sep 2022 – Feb 2026
University of Calgary *Calgary, Canada*

B.Sc. in Computer Engineering GPA: **3.69**/4.0 Sep 2016 – Feb 2021
Sharif University of Technology *Tehran, Iran*

Technical Skills

Programming: Python, C++, JavaScript, SQL
Backend / Engineering: Django, REST APIs, PostgreSQL, Node.js, Docker, Git, Linux, AWS, GitHub Actions
Data Processing: NumPy, Pandas, SciPy, ETL-style pipelines, data preprocessing, structured tensor files, large datasets
ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, model training, model evaluation, LLM workflows
Geospatial / Automation: GDAL, Rasterio, SNAP, Google Earth Engine API, Streamlit, n8n, OpenClaw, SearXNG

Work Experience

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, and retrieval of Earth observation data.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary *Calgary, Canada*

- **Developing Big Data Applications**: Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale**: Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems**: Assisted +75 students with SQL, database design, course projects, grading, and exam support.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy *Remote, United States*

- Mentored an international pod of ∼15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Research and Technical Projects

Agentic Job Search Automation Platform May 2026 – Present
OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.

- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

Spatiotemporal Earth Observation Data System Using DGGS Sep 2022 – Feb 2026

First-author publication | *M.Sc. thesis* | *C++, Python, DGGS, Wavelets, B-splines, Time-Series*

- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.
- Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.
- Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

DEM Super-Resolution Framework May 2024 – Feb 2026

Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for $4\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide.
- Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. doi:10.3390/rs17040570

Certificates

Supervised Machine Learning: Regression and Classification,
DeepLearning.AI and Stanford University, Coursera Oct 2024

Applied Machine Learning in Python,
University of Michigan, Coursera Jul 2020

Languages

English: Full Professional Proficiency (TOEFL **115**/120)
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate 2024/2025
Department of Computer Science, University of Calgary — awarded for outstanding teaching support (*link*)

Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) May 2016
Top 0.001% among all participants